A Course-End Project Report

on

“Data Science Case Studies: Titanic Survival Prediction and Bank Marketing Classification”

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by

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CASE STUDY 3

Predicting Titanic Passenger Survival using Logistic Regression

Exploratory analysis, statistical testing, model evaluation and Streamlit deployment

3.1 Problem Statement

On 15 April 1912 the RMS Titanic sank after striking an iceberg; more than 60% of the passengers in this dataset died. The objective is to build a **logistic regression** model that estimates each passenger's probability of survival from information recorded at boarding (sex, age, ticket class, fare, family aboard, port and cabin), evaluate it on unseen passengers using accuracy, precision, recall, F1-score and ROC-AUC, interpret which factors genuinely determined survival, and deploy the model as a Streamlit application. Logistic regression was chosen because the outcome is binary and each factor's effect can be read as an **odds ratio** and tested statistically — interpretability is the main deliverable.

3.2 Dataset and Methodology

titanic.csv contains 891 passengers × 12 columns with no duplicate rows. The target *Survived* is binary (0 = died, 1 = survived). The analysis was carried out in Python (pandas, seaborn, SciPy, scikit-learn, statsmodels) with a fixed random seed (42), so every number is reproducible.

Attribute	Type	Missing	Treatment in the model
Pclass, Sex	Category	0	One-hot encoded (baselines: 1st class, female)
Age	Numeric (0.42–80 yrs)	177 (19.9%)	Median imputation + AgeMissing flag; standardised
SibSp, Parch	Count	0	Standardised
Fare	Numeric (£0–512)	0	LogFare = log(1 + Fare) — skewness 4.79 → 0.39; standardised
Embarked	Category (C / Q / S)	2 (0.2%)	Most-frequent imputation, one-hot
Cabin	Text	687 (77.1%)	Too sparse to impute → binary HasCabin flag
PassengerId, Name, Ticket	Identifiers	0	Dropped

- **Exploration and testing:** summary statistics, group survival rates, histograms, box plots, pair plot, correlation matrix; chi-square tests (with Cramér's V) for categorical attributes and Mann–Whitney U tests for the non-normal numeric attributes (α = 0.05).
- **Leakage-safe preprocessing:** imputation, encoding and scaling sit inside a scikit-learn *Pipeline*, so all statistics are learned from training data only; 9 attributes become 11 model inputs.
- **Model and evaluation:** stratified 80/20 split (712 / 179 passengers); logistic regression (L2, C = 1.0, threshold 0.5); 5-fold cross-validation; accuracy, precision, recall, F1, ROC-AUC, confusion matrix and ROC curve on the test set.
- **Interpretation and deployment:** odds ratios; statsmodels *Logit* for p-values, confidence intervals and a joint Wald test; the fitted pipeline is saved with joblib and served by a Streamlit app.

3.3 Exploratory Analysis and Statistical Findings

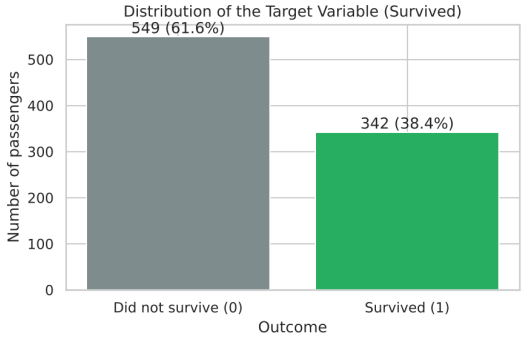


Figure 3.1 — Distribution of the target variable.

Interpretation. 549 passengers (61.6%) died and 342 (38.4%) survived. Always predicting 'did not survive' would already be 61.6% accurate, so precision, recall, F1 and ROC-AUC are reported alongside accuracy, and the split is stratified.

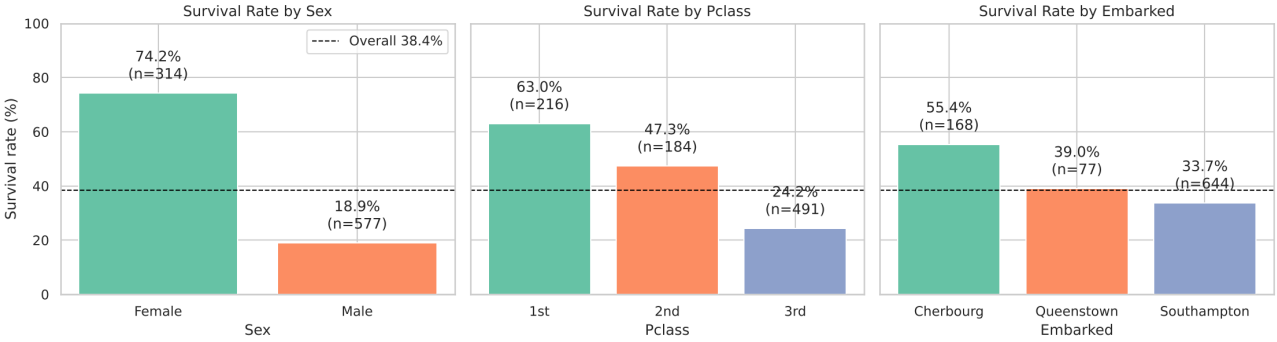


Figure 3.2 — Survival rate by sex, passenger class and port (dashed line: overall rate 38.4%).

Interpretation. Sex is the strongest single signal: 74.2% of women survived versus 18.9% of men ('women and children first'). Class shows a clear gradient — 63.0% (1st), 47.3% (2nd), 24.2% (3rd). Cherbourg passengers survived more often (55.4%) than Southampton passengers (33.7%), largely because more of them travelled 1st class.

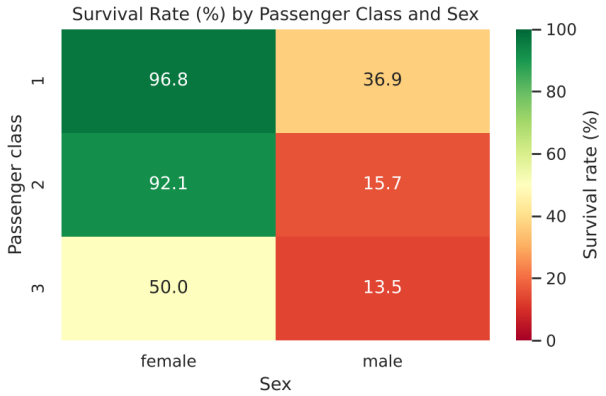


Figure 3.3 — Survival rate (%) by class and sex.

Interpretation. Sex and class act together: women in 1st and 2nd class survived at 96.8% and 92.1%, but in 3rd class only at 50.0%; men in 2nd and 3rd class fared worst (15.7% and 13.5%).

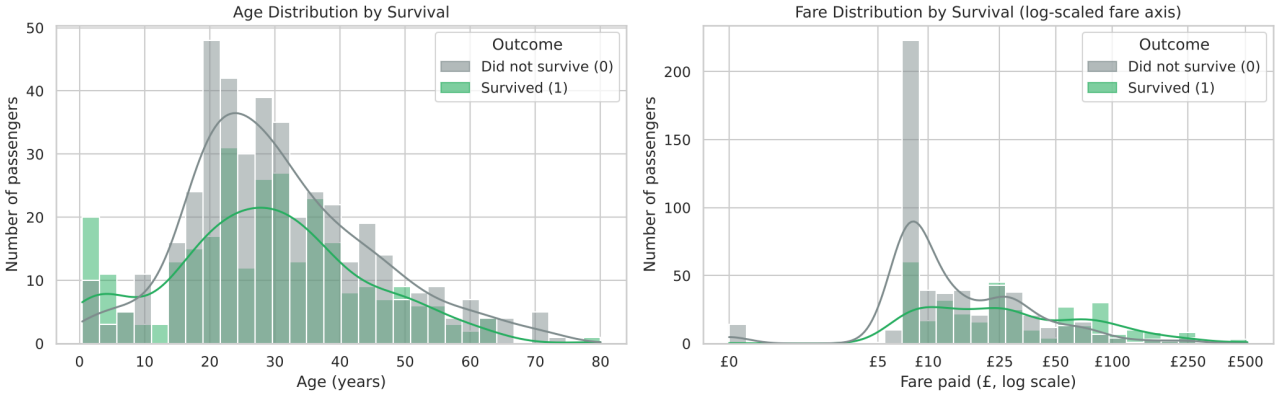


Figure 3.4 — Histograms of Age and Fare (log-scaled fare axis) by survival outcome.

Interpretation. Age distributions overlap except among young children, where survivors dominate (under-13s survived at 58.0% vs 36.7% for others). Non-survivors cluster at cheap fares of £7–£15, while above £50 survivors outnumber non-survivors (109 vs 52). Fare's long right tail motivates the log transform.

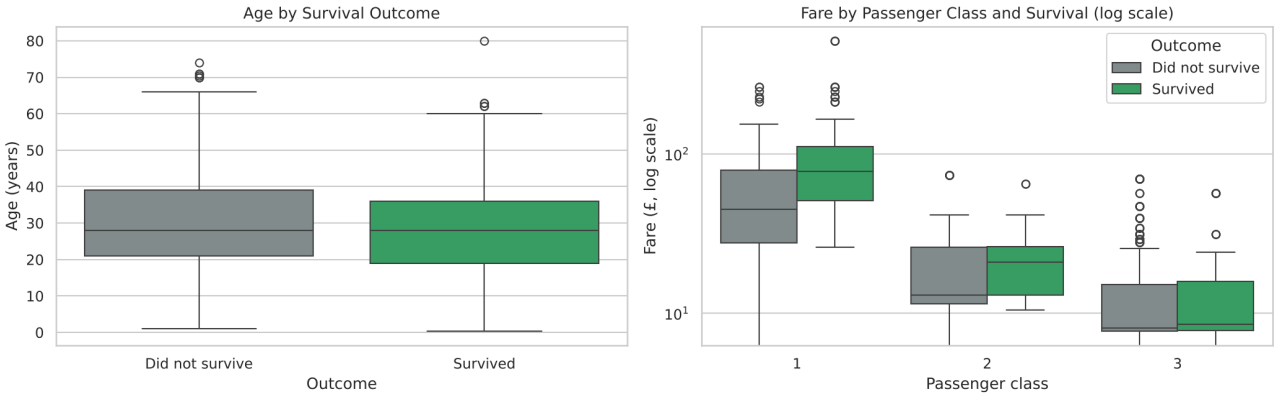


Figure 3.5 — Box plots: Age by outcome; Fare by class and outcome (log scale).

Interpretation. Median age is 28 in both groups, so age alone barely separates outcomes. Within 1st and 2nd class, survivors paid clearly more than non-survivors, so fare carries information beyond class. High-fare outliers are genuine luxury tickets and were kept.

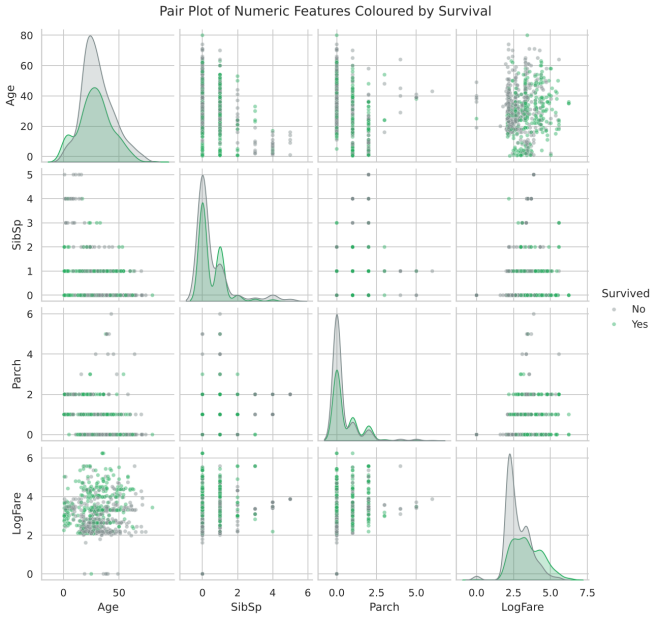


Figure 3.6 — Pair plot of numeric features coloured by survival.

Interpretation. No pair of numeric features separates the classes cleanly — survivors shift only towards higher LogFare; SibSp and Parch move together and very large families rarely survived. The categorical attributes carry most of the signal.

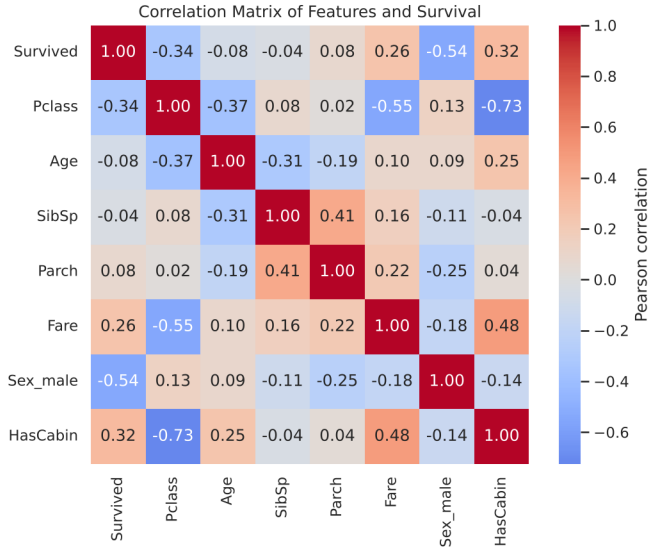


Figure 3.7 — Pearson correlation matrix (Sex_male and HasCabin as 0/1).

Interpretation. Strongest correlates of survival: Sex_male (−0.54), Pclass (−0.34), HasCabin (+0.32), Fare (+0.26); Age is almost uncorrelated (−0.08) but is correlated with class (−0.37). Pclass–HasCabin (−0.73) and Pclass–Fare (−0.55) overlap strongly, which makes their individual coefficients less certain (Section 3.5).

Statistical tests (α = 0.05)

Attribute	Test	Statistic	p-value	Effect size	Significant?
Sex	Chi-square (df 1)	260.72	1.2×10^{-58}	Cramér's V 0.541 (large)	Yes
Pclass	Chi-square (df 2)	102.89	4.6×10^{-23}	Cramér's V 0.340	Yes
HasCabin	Chi-square (df 1)	87.94	6.7×10^{-21}	Cramér's V 0.314	Yes
Embarked	Chi-square (df 2)	26.49	1.8×10^{-6}	Cramér's V 0.173	Yes
Fare	Mann–Whitney U	129,951.5	4.6×10^{-22}	Median £26.0 vs £10.5	Yes
Age	Mann–Whitney U	57,682.0	0.160	Median 28 vs 28	No

Findings. Sex, class, cabin record, port and fare are significantly associated with survival; sex has by far the largest effect. **Age is not significant on its own (p = 0.16)** — but Section 3.5 shows it becomes a significant negative factor once class is held constant, because older passengers' greater likelihood of travelling 1st class masked the effect.

3.4 Model Evaluation

Metric	Test set	5-fold CV (mean ± SD)	Meaning
Accuracy	0.793	0.799 ± 0.013	Share classified correctly (baseline 0.615)
Precision	0.767	0.753 ± 0.037	Of predicted survivors, share who survived
Recall	0.667	0.714 ± 0.038	Of actual survivors, share found
F1-score	0.713	0.732 ± 0.015	Harmonic mean of precision and recall
ROC-AUC	0.833	0.864 ± 0.020	Chance a random survivor is ranked above a random non-survivor

Explanation. The model classifies 79.3% of unseen passengers correctly — about 18 points above the 61.5% baseline. When it predicts survival it is right 76.7% of the time, and it finds two-thirds of actual survivors. ROC-AUC 0.833 indicates good discrimination from only nine raw inputs. Training accuracy (0.808), test accuracy and the cross-validated scores agree, so the model is **not over-fitting**.

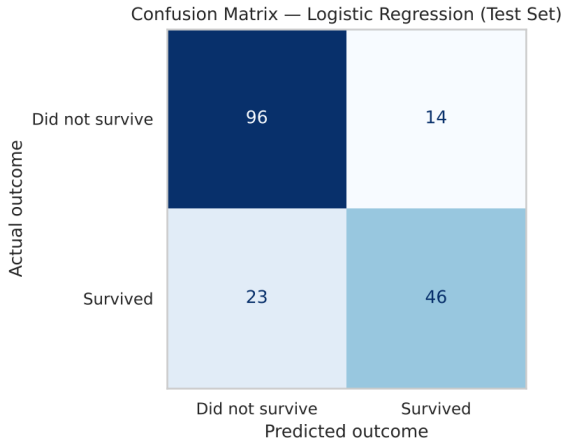


Figure 3.8 — Confusion matrix on the test set (threshold 0.5).

Interpretation. Of 110 non-survivors, 96 are correct and 14 are wrongly predicted to survive; of 69 survivors, 46 are found and 23 missed. The model errs towards predicting death, the majority class; missed survivors are mostly profiles such as adult 3rd-class men that normally implied death.

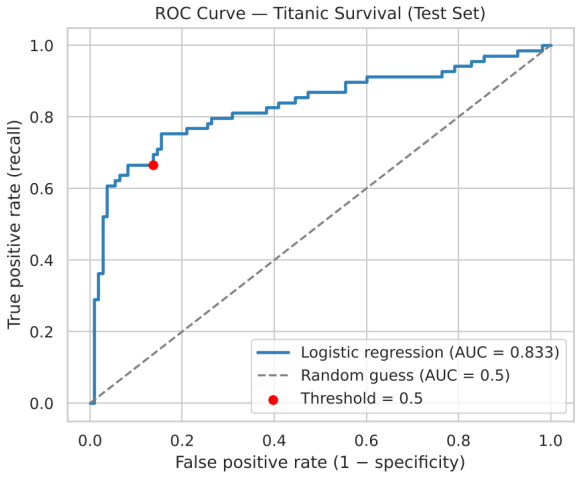


Figure 3.9 — ROC curve on the test set (red point: threshold 0.5).

Interpretation. The curve lies well above random guessing (AUC 0.833). At threshold 0.5 the model catches about 67% of survivors while wrongly flagging about 13% of non-survivors. F1 stays between 0.71 and 0.75 for any threshold from 0.30 to 0.60, so the model is not fragile to the exact cut-off; the threshold was not re-tuned on the test set, to avoid leaking test information.

3.5 Interpretation of the Model

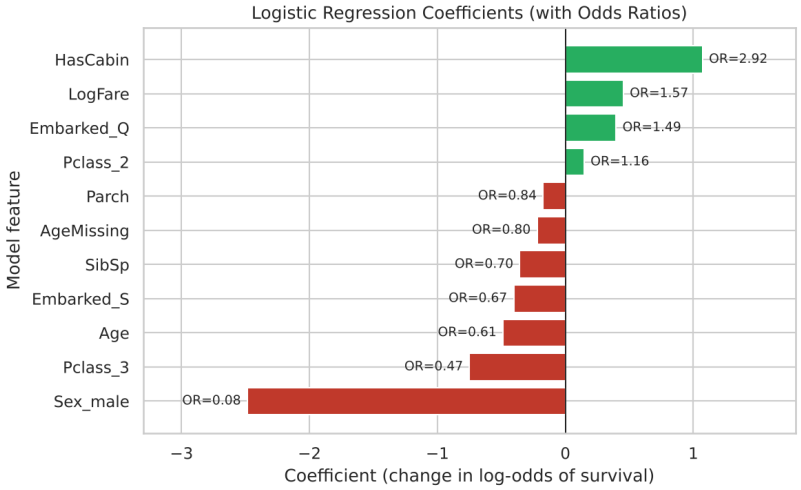


Figure 3.10 — Logistic regression coefficients (change in log-odds) with odds ratios (OR).

Interpretation. Holding other features constant, being **male** multiplies the odds of survival by 0.08 — about **12× lower** odds than a comparable woman. A recorded **cabin** multiplies the odds by 2.9 and **3rd class** roughly halves them (OR 0.47). Each SD of **age** (≈ 13 years) lowers the odds by about 39% and each SD of **siblings/spouses** by 30%, while each SD of **log-fare** raises them by 57%. Port, parents/children and the age-missing flag have small effects.

Statistical significance of each feature (statsmodels Logit, training data)

Feature	Coefficient	p-value	Odds ratio	95% CI of OR	Significant?
Sex_male	−2.623	< 0.001	0.073	0.047 – 0.113	Yes
Age (per SD)	−0.510	< 0.001	0.601	0.479 – 0.753	Yes
HasCabin	+1.158	0.002	3.184	1.521 – 6.667	Yes
SibSp (per SD)	−0.366	0.010	0.694	0.525 – 0.917	Yes
LogFare (per SD)	+0.435	0.020	1.544	1.070 – 2.229	Yes
Parch (per SD)	−0.186	0.102	0.831	0.665 – 1.037	No
Pclass_3	−0.819	0.110	0.441	0.161 – 1.205	No (jointly yes)
Embarked_S	−0.406	0.142	0.667	0.388 – 1.145	No
Embarked_Q	+0.471	0.306	1.601	0.650 – 3.943	No
AgeMissing	−0.248	0.408	0.780	0.433 – 1.405	No
Pclass_2	+0.133	0.776	1.142	0.458 – 2.846	No (jointly yes)

Explanation. Sex, age, cabin record, siblings/spouses and fare are individually significant (odds-ratio intervals exclude 1). The class dummies are not individually significant because class overlaps with cabin and fare (r = −0.73 and −0.55), widening their intervals; a **joint Wald test** gives $\chi^2 = 11.22$ (df = 2, p = 0.004), so **class does matter** — its effect is shared with the correlated variables. Port and parents/children add no reliable information. The model is highly significant (likelihood-ratio $p \approx 2.5 \times 10^{-67}$; McFadden pseudo-$R^2 = 0.364$). **Age**, not significant alone, is strongly significant here ($p < 0.001$): controlling for class removes the confounding that hid it.

Predicted survival for example passengers

Passenger profile (no family aboard unless stated)	P(survive)
Woman, 30, 1st class, fare £80, cabin recorded, Cherbourg	0.967
Man, 30, 1st class, fare £80, cabin recorded, Cherbourg	0.708
Woman, 25, 3rd class, fare £8, no cabin, Southampton	0.578
Man, 25, 3rd class, fare £8, no cabin, Southampton	0.102
Boy, 6, 2nd class, fare £30, 1 sibling + 2 parents, Southampton	0.322

With identical tickets the woman's probability far exceeds the man's in both classes. The examples also expose a limitation: the 1st-class man's 0.71 is well above the 36.9% observed for 1st-class men, because logistic regression adds effects in log-odds and cannot learn that the male penalty was harsher in practice; similarly the 6-year-old boy gets only 0.32 because age enters linearly.

3.6 Deployment with Streamlit

The complete fitted pipeline (imputation, encoding, scaling and the model) is saved as *titanic_logreg_pipeline.joblib*. The Streamlit app *app.py* loads it, collects a passenger profile through a form (sex, class, age or 'age unknown', family aboard, fare, port, cabin recorded) and displays the predicted survival probability and outcome. Because the saved object includes the preprocessing, user inputs receive exactly the same transformations as the training data. The app was tested for known-age and unknown-age passengers; it runs with `streamlit run app.py` and can be deployed from the GitHub repository on Streamlit Community Cloud.

3.7 Conclusion

- **Performance:** 79.3% accuracy, 0.767 precision, 0.667 recall, 0.713 F1 and 0.833 ROC-AUC on unseen passengers (CV AUC 0.864 ± 0.020), well above the 61.5% baseline and without over-fitting.
- **Determinants of survival:** sex was decisive (≈ 12× lower odds for men), followed by the class/cabin/fare group, age and siblings/spouses aboard; port and parents/children were not significant once these were controlled for.
- **Insight:** age looked irrelevant in a one-variable test but was significant after controlling for class — confounded features can hide real effects.
- **Limitations:** small dataset (891 rows); 20% of ages imputed with one median; additive log-odds effects only partly capture interactions such as the weaker benefit of being female in 3rd class; situational factors (location at impact, lifeboat access) are not recorded. A name-title feature, age bands and a sex × class interaction would address these.

References: Titanic dataset (titanic.csv, supplied with the case study; the Kaggle 'Titanic — Machine Learning from Disaster' training set). Pedregosa, F. et al. (2011), Scikit-learn: Machine Learning in Python, *JMLR* 12, 2825–2830. Seabold, S. & Perktold, J. (2010), statsmodels, *Proc. 9th Python in Science Conf.*

CASE STUDY 6

Comparative Analysis of Classification Algorithms for Bank Marketing Prediction

Logistic Regression, Decision Tree, K-Nearest Neighbours and Random Forest

6.1 Problem Statement

A Portuguese bank sells term deposits through telephone campaigns. Most calls fail, so calling every customer wastes agent time. The task is to **predict, before a call is made, whether a customer will subscribe**, by developing four classifiers — Logistic Regression, Decision Tree, K-Nearest Neighbours (KNN) and Random Forest — comparing them with accuracy, precision, recall, F1-score, confusion matrices and ROC-AUC, and recommending the most suitable model for targeting the bank's campaigns.

6.2 Methodology

- **Data:** UCI Bank Marketing, *bank-additional-full.csv* — 41,188 calls (May 2008 – Nov 2010), 20 attributes + target, semicolon-delimited. Chosen over the other supplied files because it contains every call (not a 10% sample) and five macro-economic indicators.
- **Attributes:** 10 numerical (age, campaign, pdays, previous, duration, emp.var.rate, cons.price.idx, cons.conf.idx, euribor3m, nr.employed) and 10 categorical (job, marital, education, default, housing, loan, contact, month, day_of_week, poutcome); target *y* (yes / no).
- **Data quality:** no blank cells, but missing information is coded as '*unknown*' (default 20.9%, education 4.2%, housing and loan 2.4%, job 0.8%, marital 0.2%) and as *pdays = 999* ('never contacted', 96.3%). 12 exact duplicate rows were removed (41,176 rows remain).
- **Preprocessing:** 'unknown' kept as its own category (dropping 21% of rows or imputing a mode would discard or invent information); pdays replaced by a binary *prev_contacted* flag (999 is a code, not a distance); categorical attributes one-hot encoded; numerical attributes standardised — essential for KNN distances. **duration was excluded** because it is only known after the call (UCI warning). Result: 19 inputs → 62 model columns.
- **Split:** 80:20 stratified train/test split (32,940 / 8,236 rows, both 11.27% positive), random_state = 42, **identical for all four models**. All preprocessing is fitted on training data only, inside a pipeline.
- **Fair tuning:** each model received a small grid search with the same 5-fold stratified cross-validation on the training set, selected by ROC-AUC (threshold-independent, robust to imbalance). The test set was used only for the final comparison.
- **Evaluation:** accuracy, precision, recall, F1, confusion matrices, ROC and precision–recall curves, a paired bootstrap test of AUC differences, threshold selection on training out-of-fold predictions, and a cumulative gains analysis.

6.3 Exploratory Analysis and Statistical Findings

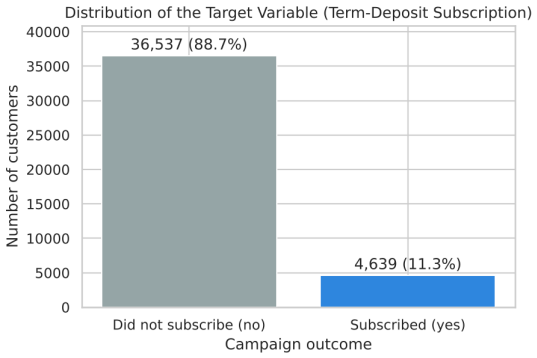


Figure 6.1 — Target-class distribution.

Interpretation. Only **4,639 of 41,176 customers (11.27%) subscribed**; 36,537 (88.73%) did not. A model that always predicts 'no' is therefore 88.7% accurate while finding no subscribers, so accuracy cannot be the basis for model selection.

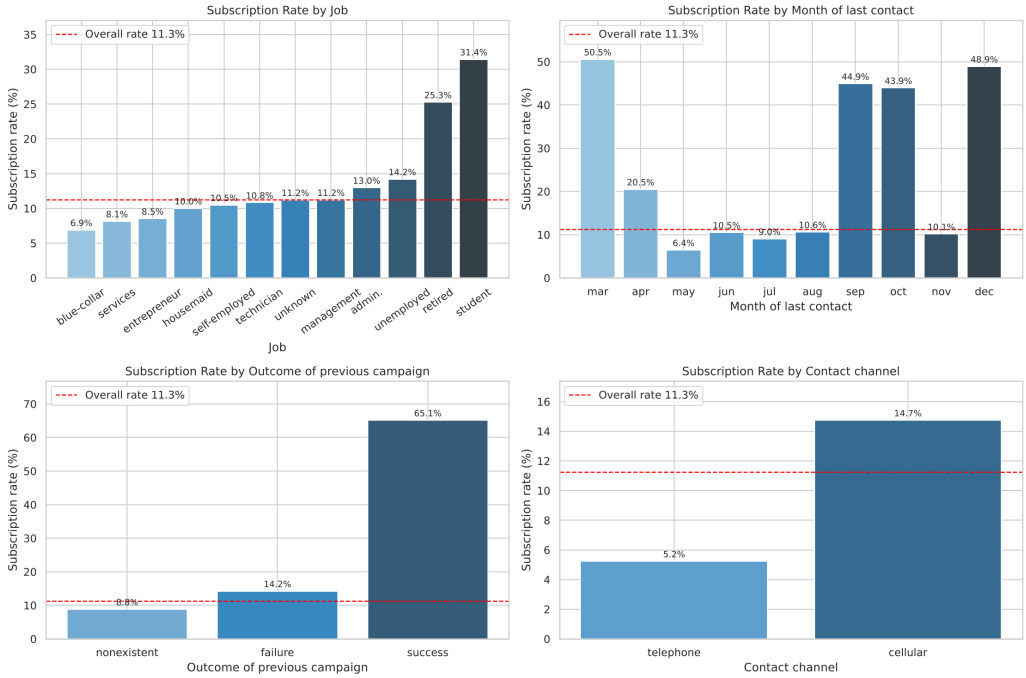


Figure 6.2 — Subscription rate by job, month, previous-campaign outcome and contact channel (red line: overall 11.3%).

Interpretation. Customers whose **previous campaign succeeded subscribe at 65.1%** (vs 14.2% after a failure and 8.8% if never contacted). March, September, October and December convert at 44–51%, while **May — a third of all calls — converts at only 6.4%**. Students (31.4%) and retirees (25.3%) are the most receptive; blue-collar workers (6.9%) the least. Mobile contacts convert at 14.7% versus 5.2% for landlines.

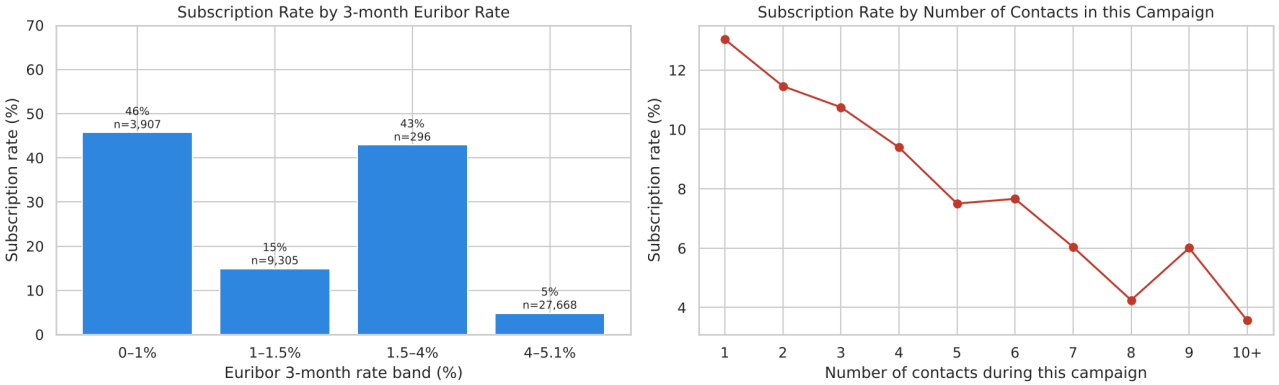


Figure 6.3 — Subscription rate by 3-month Euribor rate band and by number of contacts in the campaign.

Interpretation. With the Euribor rate below 1%, **45.7%** of customers subscribed; at 1–1.5% the rate was 14.9%; above 4% — where two-thirds of calls were made — only about **5%** did (the small 1.5–4% band of 296 calls is an exception at 42.9%). Subscribers' median Euribor rate is 1.27% versus 4.86% for non-subscribers. Conversion is highest at the first contact (≈ 13%) and falls with every repeat call (≈ 3.5% at 10 or more calls).

Statistical tests (selected; α = 0.05)

Attribute	Test	p-value	Effect size	Result
nr.employed	Mann–Whitney U	$< 10^{-300}$	0.498 (rank-biserial)	Significant — largest effect
euribor3m	Mann–Whitney U	$< 10^{-300}$	0.487	Significant
emp.var.rate	Mann–Whitney U	$< 10^{-300}$	0.433	Significant
poutcome	Chi-square (df 2)	$< 10^{-300}$	0.321 (Cramér's V)	Significant
month	Chi-square (df 9)	$< 10^{-300}$	0.275	Significant
job / contact	Chi-square	$\approx 10^{-199}$ / 10^{-189}	0.153 / 0.145	Significant
age	Mann–Whitney U	0.016	0.022	Significant but negligible
housing	Chi-square (df 2)	0.057	0.012	Not significant
loan	Chi-square (df 2)	0.577	0.005	Not significant

With 41,176 records almost any difference is 'significant', so effect size is the meaningful measure. The economic indicators and previous-campaign outcome have the largest effects; housing and personal loans have no detectable relationship with subscription.

Three evidence-based observations.

- **Subscription is rare but concentrated:** 11.3% overall, yet 65.1% among previous-campaign successes and 25–31% among retirees and students.
- **The economic environment dominates:** conversion swings from about 5% to 46% with the interest-rate level, and the macro indicators show the largest effect sizes of all attributes.
- **More calls do not mean more sales:** conversion falls steadily with every repeat contact, and mobile contacts convert almost 3× better than landline ones.

6.4 Model Development and Performance Comparison

Model	Tuned hyper-parameters	CV ROC-AUC (train)
Logistic Regression	C = 0.1	0.790
Decision Tree	max_depth = 8, min_samples_leaf = 20	0.785
KNN	n_neighbors = 101	0.784
Random Forest	300 trees, max_depth = 12, min_samples_leaf = 5	0.799

Final performance comparison table (test set, 8,236 customers, threshold 0.5)

Model	Acc.	Prec.	Recall	F1	ROC-AUC	PR-AUC	TP	FN	FP
Logistic Regression	0.899	0.663	0.210	0.319	0.800	0.453	195	733	99
Decision Tree	0.897	0.590	**0.282**	**0.382**	0.805	0.458	262	666	182
KNN	0.899	**0.668**	0.208	0.317	0.795	0.450	193	735	96
Random Forest	**0.900**	0.661	0.233	0.344	**0.811**	**0.483**	216	712	111
Random Forest, threshold 0.21	0.878	0.468	**0.601**	**0.526**	0.811	0.483	558	370	634

Baselines: accuracy 0.887 (always 'no'); PR-AUC 0.113 (random ranking). 928 actual subscribers in the test set.

Explanation. All four models reach ≈ 0.90 accuracy — barely one point above the 0.887 baseline — so accuracy cannot separate them. At the default 0.5 threshold every model misses most subscribers (recall 0.21–0.28) because, with 11% positives, few customers ever reach a 50% predicted probability. Precision of 0.59–0.67 means a flagged customer is 5–6× more likely to subscribe than a random one. Random Forest has the best ranking quality (ROC-AUC 0.811, PR-AUC 0.483); the Decision Tree's higher F1 at 0.5 is a threshold effect from its coarser probabilities, not better discrimination.

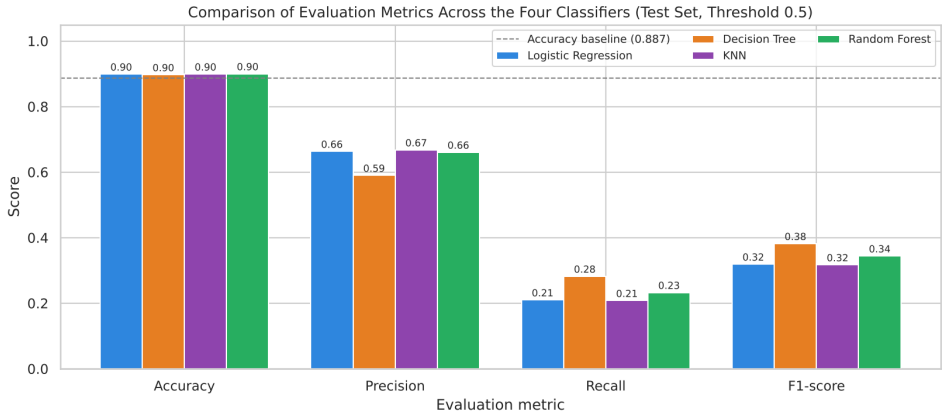


Figure 6.4 — Grouped bar chart of accuracy, precision, recall and F1 for the four classifiers (threshold 0.5).

Interpretation. The accuracy bars are indistinguishable and sit just above the baseline line, while recall and F1 reveal both the differences between models and their shared weakness. Choosing by accuracy alone would treat the models as equal and hide that each misses three-quarters of subscribers at the default threshold.

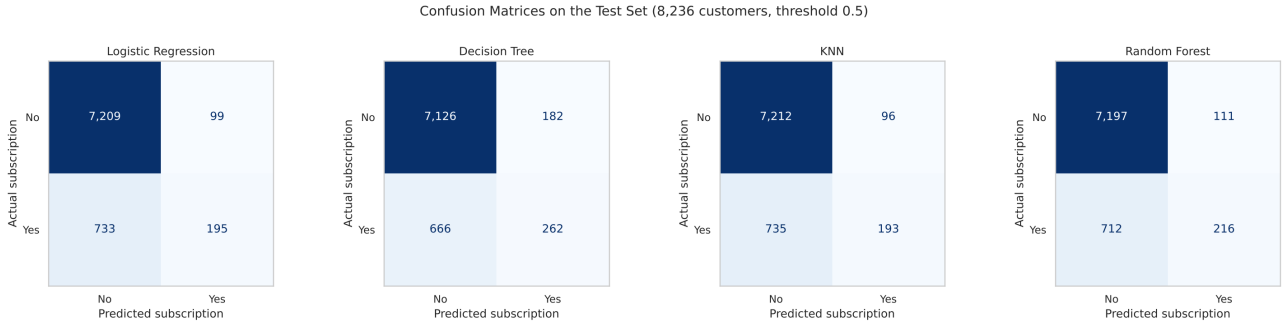


Figure 6.5 — Confusion matrix for each classifier on the test set (threshold 0.5).

Interpretation. False negatives dominate the errors: every model misses 666–735 of the 928 subscribers — each one a lost sale. False positives (96–182 of 7,308 non-subscribers) only cost a wasted call. The Decision Tree finds the most subscribers (262) with the most false positives (182); Logistic Regression and KNN are the most conservative (195 and 193 found). Because a missed subscriber is far costlier than an extra call, the 0.5 threshold is too strict for this business.

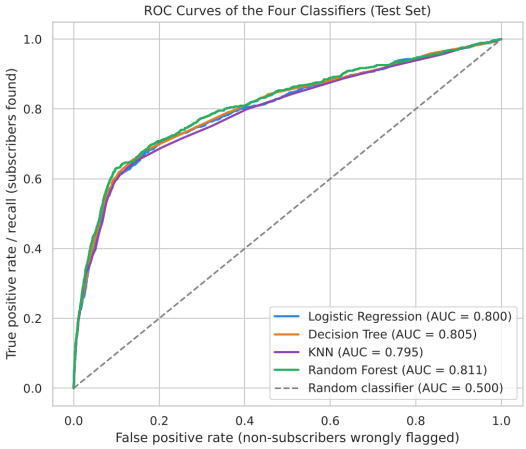


Figure 6.6 — ROC curves of the four classifiers with ROC-AUC.

Interpretation. All curves lie well above the diagonal and close together; Random Forest (AUC 0.811) is on or above the others over most of the range. About 60–65% of subscribers can be found while flagging only ~10% of non-subscribers; beyond that elbow every curve flattens, because the remaining subscribers look like non-subscribers on pre-call data. No algorithm breaks this ceiling — the limit is the information available, not the model.

Is the best model reliably better?

A paired bootstrap (1,000 resamples of the test set) estimates the 95% confidence interval of Random Forest's AUC advantage: **+0.011 over Logistic Regression (0.004 to 0.018)** and **+0.016 over KNN (0.008 to 0.025)** — both reliable — but only **+0.006 over the Decision Tree (−0.0003 to 0.014)**, which is borderline and not significant at the 5% level (RF better in 96.9% of resamples). Random Forest is the best ranker, by a modest margin.

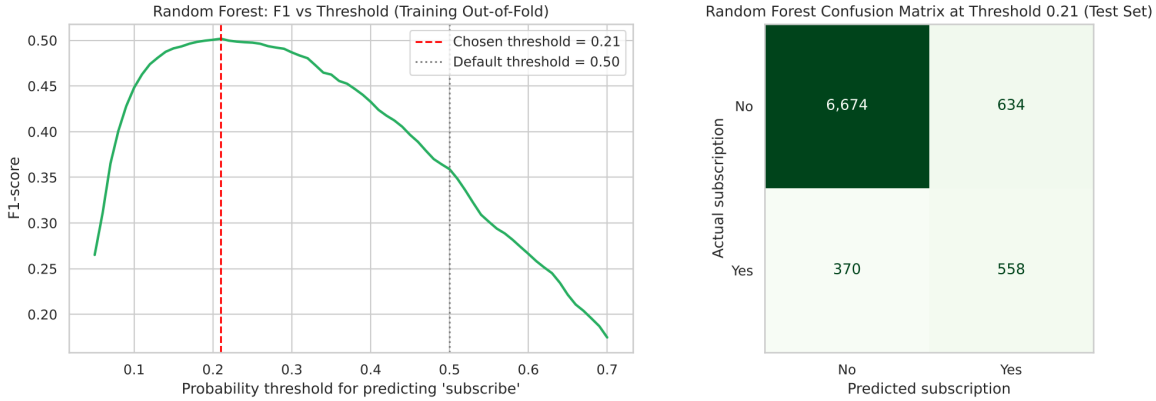


Figure 6.7 — Random Forest: F1 versus threshold on training out-of-fold predictions (left) and test confusion matrix at the chosen threshold (right).

Interpretation. The F1-maximising threshold, found on training data only, is **0.21**. Applied once to the test set, Random Forest now **finds 558 of 928 subscribers (recall 0.601, up from 0.233) with precision 0.468** — almost half of flagged customers subscribe, 4.2× the base rate — and F1 rises from 0.344 to 0.526. It flags 14.5% of customers: 523 extra calls buy 342 extra sales. Accuracy falls to 0.878 even though the model is far more useful, a direct illustration of why accuracy misleads on imbalanced data.

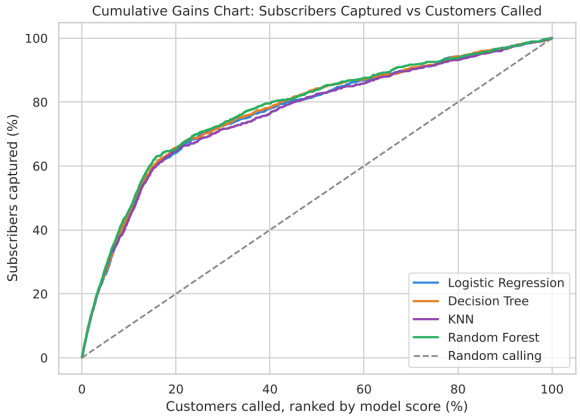


Figure 6.8 — Cumulative gains: share of subscribers reached when calling customers in model-ranked order.

Interpretation. Calling in Random Forest's order reaches **46.2% of all subscribers from the top 10% of customers** (a 4.6× lift over random calling), 65.3% from the top 20% and 73.3% from the top 30%. The four models give similar curves; Random Forest leads in the top decile, where call budgets are tightest.

6.5 Best Model, Business Recommendations and Limitations

Best model: Random Forest with a 0.21 decision threshold. It has the highest ROC-AUC (0.811) and PR-AUC (0.483), is reliably better than Logistic Regression and KNN, and at its tuned threshold gives the best balance of the metrics that matter here (recall 0.601, precision 0.468, F1 0.526). It suits the problem because the bank needs a *ranked call list* with adjustable cut-off, and because a forest of trees handles the strongly correlated economic indicators (r up to 0.97) and their interactions with month and contact history without manual engineering, while averaging 300 trees makes it more stable than a single tree. Its importances agree with the EDA: economic indicators account for ≈ 48% of its splitting power and past-campaign history for ≈ 23%. Trade-off: it is less transparent than Logistic Regression (AUC 0.800), the best fallback if coefficient-level explanation is required.

Business recommendations

- **Score every customer before a campaign and call in ranked order.** Calling the top ≈ 15% (threshold 0.21) reaches about 60% of likely subscribers, with nearly half of calls converting instead of 11%. For a fixed budget, the gains curve gives the share of subscribers reachable.
- **Prioritise high-propensity segments and cap repeat calls:** previous-campaign successes, students, retirees and mobile contacts first; limit attempts per customer (about 3), since conversion falls with every repeat.
- **Adapt campaign intensity to economic conditions and retrain regularly**, because propensity swings between ≈ 5% and ≈ 46% with the interest-rate environment.

Limitations

- **Time dependence:** the data covers the 2008–2010 crisis and the economic indicators are identical for all customers on a given day, so the model partly learns *when* calls succeed. The random split mixes time periods; a time-ordered split would be stricter and would probably show lower performance, so the model must be revalidated before use in new economic conditions.
- **Imbalance and unknown costs:** the 0.21 threshold maximises F1, which weights precision and recall equally. The real cost of a call and profit of a deposit are not in the data; with them, the threshold should be set to maximise expected profit and could shift.

Conclusion

Four classifiers were trained on identical, leakage-free inputs and compared on the same test set. All reached ≈ 90% accuracy, barely above the 88.7% of never predicting a subscription, which shows why accuracy cannot guide the choice. Judged by ROC-AUC, PR-AUC, recall, F1 and the cost of missed subscribers, **Random Forest with a 0.21 threshold** is the most useful model: it identifies 60% of subscribers from 14.5% of customers, with nearly half of its calls converting. As a check on the required exclusion of *duration*: adding it raises Logistic Regression's AUC from 0.800 to 0.939 — an illusory gain from information that only exists after the call.

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